Fix gauge later

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{\lambda}{2}(\partial_\mu A^\mu)^2$$

Gutrent

λ variable

$$\mathcal{L} = \mathcal{L}(A, \partial A, \lambda)$$

EL $\quad \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} = \frac{\partial \mathcal{L}}{\partial A_\nu}$

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \lambda} = \frac{\partial \mathcal{L}}{\partial \lambda}$$

$\mathcal{L}$ does not depend on $\partial_\mu \lambda$ $\quad \frac{\partial \mathcal{L}}{\partial \partial_\mu \lambda} = 0$

$0 = \frac{1}{2}(\partial_\mu A^\mu)^2$

impose $\partial_\mu A^\mu = 0$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$\mathcal{L}_0 = -\frac{1}{4} F_{\mu\nu} F_{\mu\nu} \qquad \mathcal{L}_1 = \frac{\lambda}{2}(\partial_\mu A^\mu)^2$$

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} = \frac{\partial \mathcal{L}}{\partial A_\nu}$$

$$\partial_\mu \frac{\partial \mathcal{L}_0}{\partial \partial_\mu A_\nu} + \partial_\mu \frac{\partial \mathcal{L}_1}{\partial \partial_\mu A_\nu} = \cancel{\frac{\partial \mathcal{L}_0}{\partial A_\nu}}^{\,0} + \cancel{\frac{\partial \mathcal{L}_1}{\partial A_\nu}}^{\,0} = 0$$

done already

$$-\partial_\mu F^{\mu\nu}$$

$$\frac{\partial \mathcal{L}_1}{\partial \partial_\mu A_\nu} = \frac{\partial}{\partial \partial_\mu A_\nu} \frac{\lambda}{2}\left(\eta^{\alpha\beta} \partial_\alpha A_\beta\right)^2$$

$$= \lambda \left(\eta^{\alpha\beta} \partial_\alpha A_\beta\right) \frac{\partial}{\partial \partial_\mu A_\nu} \eta^{\alpha\beta} \partial_\alpha A_\beta = \eta^{\mu\nu} \lambda \, \partial_\alpha A^\alpha$$

$$\partial_\mu \frac{\partial \mathcal{L}_1}{\partial \partial_\mu A_\nu} = \eta^{\mu\nu} \partial_\mu \lambda \, \partial_\alpha A^\alpha + \lambda \eta^{\mu\nu} \partial_\mu \partial_\alpha A^\alpha$$

$$\boxed{-\partial_\mu F^{\mu\nu} + \eta^{\mu\nu} \cancel{\partial_\mu \lambda} \, \partial_\alpha A^\alpha + \lambda \eta^{\mu\nu} \partial_\mu \partial_\alpha A^\alpha} = 0$$

impose ahead of time $\quad \lambda =$ Gutent

$$-\partial_\mu F^{\mu\nu} = -(\eta^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu)A_\mu$$

$$(\lambda\eta^{\alpha\nu}\partial_\alpha\partial^\mu)A_\nu$$

$$[\eta^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu - \lambda\partial^\nu\partial^\mu]A_\nu = 0$$

$$\boxed{[\eta^{\mu\nu}\partial^2 - (1+\lambda)\partial^\mu\partial^\nu]A_\nu = 0}$$

Green's function $[\eta^{\mu\nu}\partial^2 - (1+\lambda)\partial^\mu\partial^\nu]\, G_{\nu\sigma}(x) = \delta^\mu_\sigma\, \delta(x)$

momentum space: $(-\eta^{\mu\nu}p^2 + (1+\lambda)p^\mu p^\nu)\,\hat{G}_{\nu\sigma}(p) = \delta^\mu_\sigma$

$$\hat{G}_{\nu\sigma}(p) = -\frac{\eta_{\nu\sigma} + B\, p_\nu p_\sigma}{p^2} \quad \text{ansatz} \quad (B \text{ arbitrary})$$

$$(-\eta^{\mu\nu}p^2 + (1+\lambda)p^\mu p^\nu)\left(\frac{-\eta_{\nu\sigma} - B\, p_\nu p_\sigma}{p^2}\right)$$

$$= \frac{1}{p^2}\left(\delta^\mu_\sigma p^2 + p^\mu p_\sigma \cancel{p^2} B - (1+\lambda)p^\mu p_\sigma - B(\cancel{1}+\lambda)p^\mu p_\sigma p^2\right)$$

$$= \frac{1}{p^2}\left(\delta^\mu_\sigma p^2 - (1+\lambda)p^\mu p_\sigma - B\lambda\, p^\mu p_\sigma p^2\right)$$

need $\quad -(1+\lambda) - B\lambda p^2 = 0 \qquad B = -\frac{(1+\lambda)}{\lambda p^2} = -\left(1+\frac{1}{\lambda}\right)\frac{1}{p^2}$

$$\boxed{\hat{G}_{\mu\nu}(p) = -\frac{\eta_{\mu\nu} - \left(1+\frac{1}{\lambda}\right)\frac{p_\mu p_\nu}{p^2}}{p^2}}$$

freedom to set λ to what we want

Feynman gauge $\quad \lambda = -1$

Quantization of fermions

$\bar\Psi \gamma^0 \partial_0 \Psi + \dots$

$\gamma^0 = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}$

$$\mathcal{L} = \frac{i}{2}\left(\bar\Psi \gamma^\mu \partial_\mu \Psi - \partial_\mu \bar\Psi \gamma^\mu \Psi\right) - m \bar\Psi \Psi$$

$\bar\Psi = \Psi^* \gamma^0$

$$\Pi = \frac{\delta \mathcal{L}}{\delta \dot\Psi} = \frac{i}{2}\bar\Psi \gamma^0 = \frac{i}{2}\Psi^* (\gamma^0)^2 = \frac{i}{2}\Psi^*$$

quantize: $\{\Psi^\alpha(x,t), \Pi_\beta(x',t)\} = i\,\delta^\alpha_\beta\,\delta(x,x')$

$\rightarrow$ anticommutator $\quad \{A,B\} = AB + BA$

---

eventually we write Ψ as a Fourier expansion with fields $b(k)$, $b^+(k)$

with $\{b(k), b^+(k')\} = \delta(k-k')$ $\quad \| \quad \{b(k), b(k')\} = 0 \quad \{b^+(k), b^+(k')\} = 0$

$|k\rangle = b^+(k)|0\rangle$ state with one electron with momentum k

should be

$b^+(k')|k\rangle = 0$ if $k = k'$

$\|$

$b^+(k')\,b^+(k)|0\rangle = -b^+(k)\,b^+(k')|0\rangle$

$b(k)\,b(k') + b(k')\,b(k) = 0$

$\rightarrow b(k)\,b(k') = -b(k')\,b(k)$

$$\| b^+(k')|k\rangle\|^2 = \langle k| b(k')\, b^+(k')\,|k\rangle$$

$$= \langle k| b(k')\, b^+(k')\, b^+(k)|0\rangle$$

$$= -\langle 0| b(k)\, b(k')\, b^+(k)\, b^+(k')|0\rangle$$

$$= -\langle 0| b(k)\left(-b^+(k)\,b(k') + \delta(k-k')\right) b^+(k')|0\rangle$$

$$= \langle 0| b(k)\, b^+(k)\, b(k')\, b^+(k')|0\rangle - \delta(k-k)\langle 0| b(k)\, b^+(k')|0\rangle$$

$$= \langle 0| b(k)\, b^+(k)\left(-b^+(k')\,b(k') + \delta(\quad\right.|0\rangle$$

$b(k')\,b^+(k')$

$\{A,B\} = C$

then $AB = -BA + C$

$$\{\psi(x,t), \psi(x',t)\} = 0 \qquad E_k = \sqrt{k^2+m^2}$$

start with $$\psi_\alpha(x,0) \propto \int \frac{d^3k}{(2\pi)^{3/2}} \left(A(k)\, e^{-ik\cdot x} + B(k)\, e^{ik\cdot x} \right)$$

$$\psi_\alpha(x,0) \propto \int \frac{d^3k}{(2\pi)^{3/2}} \frac{1}{(E_k/m)^{\frac{1}{2}}} \sum_{s=\pm 1} \left(b(k,s)\, u_\alpha(k,s)\, e^{-ikx} + d^\dagger(k,s)\, v_\alpha(k,s)\, e^{ik\cdot x} \right)$$

$b^\dagger(k,s)$ creates particle with left orientation, spin s, and momentum k

$$\{b(k,s), b^\dagger(k',s')\} = \delta_{s,s'}\, \delta(k-k')$$

$$\{b(k,s), b(k',s')\} = 0$$

$$\{b^\dagger(k,s), b^\dagger(k',s')\} = 0$$